

SCH3U

Test : Unit 5

Gases & Atmospheric Chemistry

Name: _____

Instructions:

You will have the entire period to complete the test (75 minutes)

The test is made up of 4 sections:

Section A – Multiple Choice

Section B – True and False

Section C – Short Answer

Section D – Long Answer

Please follow all the instructions

Please read all questions carefully

A mark breakdown of the test can be found after these instructions

Good Luck!

Mark Summary

Section	Knowledge /Understanding	Thinking	Application	Communication	TOTAL
A	/10	/	/	/	10
B	/5	/	/	/	5
C	/	/3	/7	/2	12
D	/1	/2	/1	/3	7
Sub-Totals	/ 16	/ 5	/ 8	/ 5	/ 34

Comments:

Part A Multiple Choice (1 mark each – K/U)

1) What are the top four gaseous components of our atmosphere (by percentage)?

- A. Nitrogen, Oxygen, Hydrogen, Ozone
- B. Nitrogen, Oxygen, Carbon Dioxide, Helium
- C. Nitrogen, Oxygen, Argon, Hydrogen
- D. Nitrogen, Oxygen, Argon, Carbon Dioxide

2) What is the correct balanced chemical equation?

- A. $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$
- B. $3 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 3 \text{H}_2\text{O}(\text{g})$
- C. $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$
- D. $\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$

3) Standard conditions of temperature and pressure (STP) are taken to be?

- A. 25°C 760 mmHg
- B. 298 K 1 atm
- C. 273 K 1 atm
- D. 0°C 76 mmHg

4) Given the units for pressure (atm), temperature (K), volume (L) and amount (mol) of a gas, what are the units for the gas constant (R) in the equation $PV = nRT$?

- A. $\text{L}^{-1} \text{atm}^{-1} \text{mol}^{-1} \text{K}^{-1}$
- B. $\text{L atm mol}^{-1} \text{K}^{-1}$
- C. $\text{L}^{-1} \text{atm mol}^{-1} \text{K}^{-1}$
- D. $\text{L atm}^{-1} \text{mol}^{-1} \text{K}^{-1}$

5) What are the three types of motion that gas molecules are capable of?

- A. Angular, Rotational, Linear
- B. Linear, Vibrational, Angular
- C. Translational, Vibrational, Rotational
- D. Rotational, Vibrational, Angular

- 6) As the temperature of a gas increases the gas molecules will increase in?
- A. Potential Energy
 - B. Kinetic Energy
 - C. Electrical Energy
 - D. Total Energy
- 7) What statement best describes Boyle's Law?
- A. The volume of a fixed amount of gas at constant pressure is directly proportional to the temperature
 - B. At a fixed temperature and pressure, the volume of a gas is directly proportional to the amount of gas
 - C. For a fixed amount of gas at a constant temperature, gas volume is inversely proportional to gas pressure
 - D. The temperature of a gas is directly proportional to the average translational kinetic energy of its molecules
- 8) A 72.8 L constant volume cylinder containing 1.85 mol of He is heated until the pressure reaches 3.50 atm. What is the final temperature in degrees Celsius?
- A. $1.41 \times 10^3 \text{ }^\circ\text{C}$
 - B. $1.41 \times 10^{-3} \text{ }^\circ\text{C}$
 - C. $1.41 \times 10^2 \text{ }^\circ\text{C}$
 - D. $1.41 \times 10^{-2} \text{ }^\circ\text{C}$
- 9) You have a fixed amount of O₂ gas with a volume of 2.0 L at constant pressure. The gas has an initial temperature of 293 K. If the gas is heated to 353 K, what will the new volume of the O₂ gas be (to the nearest decimal point)?
- A. 2.7 L
 - B. 3.0 L
 - C. 2.0 L
 - D. 2.4 L

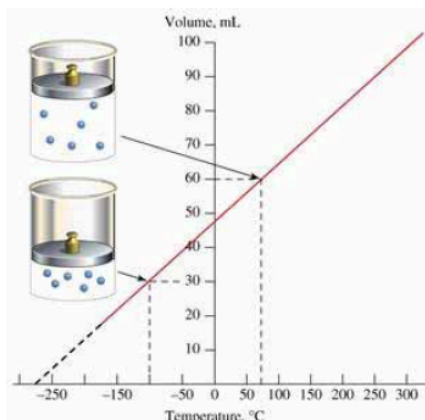
- 10) The picture on the right represents a steel tank filled with hydrogen gas at 20°C and 3 atm. Which of the diagrams below would best illustrate the tank if the temperature were lowered to -20°C (Hydrogen freezes at -259°C)?



- A.
- B.
- C.
- D.

Part B True or False (1 mark each – K/U)

- | | | |
|--|------|-------|
| 1) The kinetic energy of gas molecules is lower than liquid molecules. | True | False |
| 2) 1 mol of gas at STP will have a volume of 22.4 L. | True | False |
| 3) If the pressure of a fixed amount of gas at constant temperature is doubled the volume of that gas will also double. | True | False |
| 4) Hydrogen gas will react with oxygen gas in a 1:2 ratio to produce water (in the gaseous state). | True | False |
| 5) This graph below depicts Charles's Law; the dotted line extending from the bottom left end of the red line represents a hypothetical gas that does not freeze. Assuming that all gases are hypothetical and do not freeze they would all have the same starting point on the graph. | True | False |



Part C Short Answer (A:7 C:2 T:2)

- 1) You purchase a bag of potato chips at the beach snack stand for your picnic in the mountains. At the picnic the bag has inflated to the point of bursting. Use your knowledge of gas behaviour to explain, assume temperature is constant.
(C:2 T:2)

- 2) A 35.8 L cylinder of Ar (g) is connected to an empty 1875 L tank. If the temperature is held constant and the final pressure is 721 mmHg, what must have been the original gas pressure in the cylinder in atmospheres?

Use the G.U.E.S.S method

(A:3)

- 3) A 12.8 L cylinder contains 35.8 g O₂ at 46°C. What is the pressure of this gas, in atmospheres? Use the G.U.E.S.S. method. ($R = 0.082057 \text{ L atm mol}^{-1} \text{ K}^{-1}$)

(A:4)

Part D Long Answer (K:1 C:3 T:3 A:1)

Explain how an ordinary drinking straw in a glass of pop works in regards to pressure? You may use a diagram or words to explain.

(C:2 T:3)

Diet coke is less dense than regular coke, show the relationship between height of the straw (h) and density of the liquid (d) at constant pressure (P) and gravity (g), to show what type of pop could be pushed up a longer straw.

Brand of pop that travels higher.		[/1 _{K/U}]
Equation for height of straw using pressure, density and gravity.	$h =$	[/1 _A]
Relationship between height and density at constant pressure.		[/1 _C]